

Team & Project Registration

Title of the Project: Agri-Bridge: AI-Enabled Multilingual Circular Economy Platform for Smallholder Farmers

Team Name: Agri-Bridge

Team Lead: Prakruthi BR

Mission Tagline: “Empowering smallholder farmers to monetize crop residues, eliminate stubble burning, and access vernacular AI farm intelligence through a decentralized circular marketplace.”

Institution: Department of Computer Applications, Alliance University, Bengaluru

Team Member & Role	Registration / USN	Email Address	Contact Number
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ABSTRACT

Executive Summary & Problem-Solution Fit

PROJECT ABSTRACT (STRICT SINGLE-PARAGRAPH FORMAT):

Across rural India, **smallholder farmers face severe economic distress from opaque mandi middlemen and crop diseases, while lacking direct channels to monetize crop residues—leaving open-field stubble burning as their only viable option**, an environmental crisis discharging over 149 million tonnes of CO₂ and hazardous particulate smog annually. **Agri-Bridge addresses this dual socio-economic and ecological dilemma by establishing an AI-enabled, mobile-first circular bio-economy marketplace that directly connects small-scale farmers with industrial bioenergy, packaging, and composting buyers.** Our automated carbon accounting engine immediately calculates fair-market farmer payouts and net CO₂ emissions abated per residue listing. To dismantle technical and literacy barriers, Agri-Bridge provides hands-free floating voice navigation via the **Web Speech API across six regional languages** (Hindi, Kannada, Tamil, Telugu, Malayalam, English), alongside an interactive **computer vision AI Plant Doctor for instant disease diagnosis**, real-time APMC Mandi market tickers, and gamified carbon credit leaderboards. During the 15-hour hackathon, our team delivers a production-ready, fully demonstrable MVP that validates seamless vernacular trade, automated carbon calculations, and voice-guided farmer interaction, creating tangible rural wealth while driving regional decarbonization.

149M+ Tonnes
Stubble & Smog Addressed

6 Indian Languages
Hands-Free Voice AI

Direct B2B Market
Zero Middleman Leakage

15-Hour MVP
Fully Functional Prototype

THE SOLUTION

Transforming Agricultural Waste into Rural Wealth & Clean Energy via Three Core Pillars

1. P2P BIOMASS MARKETPLACE [CIRCULAR BIO-ECONOMY]

- **Direct B2B Stubble Trade:** Farmers list surplus crop residue (paddy straw, wheat stubble, bagasse) directly to bioenergy & packaging plants.
- **Automated Carbon Calculator:** Proprietary math instantly computes fair farmer earnings and metric tonnes of CO₂ emissions prevented from open fire.
- **Proximity-Based Routing:** Industrial buyers filter listings by GPS radius to minimize transit freight, optimize logistics, and place instant deal quotes.

2. VOICE AI & VERNACULAR UX [ZERO LITERACY BARRIER]

- **Web Speech API Navigation:** Floating microphone widget lets farmers navigate the platform verbally using natural spoken mother-tongue commands.
- **6 Regional Indian Languages:** Complete localization and UI toggle in Hindi, Kannada, Tamil, Telugu, Malayalam, and English.
- **Voice-Assisted AI Chatbot:** Converts spoken farmer queries into structured answers for crop protection, fertilizer schedules, and state subsidies.

3. AI PLANT DOCTOR & MANDI [FARM INTELLIGENCE & ESG]

- **Computer Vision Diagnostics:** Farmers snap/upload leaf photos; animated canvas scanner diagnoses diseases and prescribes organic remedies.
- **Dynamic APMC Mandi Ticker:** Live crop prices with real-time rise/fall trend indicators to protect farmers against predatory middlemen sales.
- **Carbon Credit Leaderboard:** Gamifies stubble-free agriculture, ranking top regional farmers with Eco-Champion badges and green reputation points.

LEAN CANVAS

Commercial Viability, Business Model & Sustainable Market Architecture

1. PROBLEM

- **Stubble Smog Emergency:** 149M+ tonnes crop residue burned annually causing lethal air pollution and soil damage.
- **Farmer Economic Leakage:** Smallholders lose 25-40% margin to middlemen with zero channels to monetize farm residue.
- **Digital Accessibility Gap:** Complex text-heavy apps fail rural farmers due to linguistic divide and lack of voice UX.

2. UNIQUE VALUE PROP (USP)

- **Voice-First Bio-Economy:** First voice-driven circular platform linking smallholders directly with green industries.
- **Automated Dual-Yield Engine:** Converts crop stubble into instant fair ₹ profit + certified CO₂ abatement per deal.
- **Zero Literacy Friction:** Complete 6-language vernacular localization with AI plant disease diagnostics.

3. CUSTOMER SEGMENTS

- **Supply (Farmers):** 100M+ smallholder & marginal farmers across Tier-2/Tier-3 agricultural belts in India.
- **Demand (Industries):** Biomass power plants, paper mills, bio-coal briquette manufacturers, composting units.
- **Institutional Partners:** Farmer Producer Organizations (FPOs), pollution control boards, corporate ESG buyers.

4. COST STRUCTURE

- **Cloud & Infrastructure:** Netlify CI/CD hosting, SSL security, serverless compute & edge bandwidth (~20%).
- **AI & API Services:** Web Speech API processing, computer vision model inference & language pipelines (~35%).
- **Field Operations:** FPO grassroots partnerships, farmer onboarding workshops & verification outreach (~45%).

5. UNFAIR ADVANTAGE & METRICS

- **Unfair Advantage:** Proprietary proximity routing algorithm + native voice navigation in 6 Indic dialects.
- **Core Growth Metrics:** Residue tonnage traded, CO₂ tonnes abated, average farmer net income boost (₹).
- **Community Retention:** Gamified Carbon Credit Leaderboard driving continuous post-harvest farmer engagement.

6. REVENUE STREAMS

- **Transaction Commission:** 2.5% platform brokerage fee on verified industrial biomass purchase contracts.
- **Industrial SaaS Analytics:** Subscription tier for bioenergy plants to track regional feedstock supply and radius.
- **Carbon Token Brokerage:** Aggregated verified emissions reductions monetized via voluntary carbon markets.

ARGONYX'26

EXECUTION PLAN(MVP IN 15 HRS)

System Flow Architecture, 15-Hour Agile Roadmap & Production Tech Stack

1. USER TOUCHPOINT
Farmer / Industrial Buyer
Voice Mic (Web Speech) or UI

2. MULTILINGUAL APP
React.js + Vite Client
6 Indic Languages (PWA)

3. CORE ENGINES
Carbon Math & Proximity Logic
AI Plant Doctor Diagnostic Canvas

4. MARKET OUTCOMES
B2B Deals + Carbon Credits
Live APMC Mandi Price Ticker

15-HOUR AGILE SPRINT ROADMAP (FEASIBILITY)

- **Hours 0 – 3: Architecture & Vernacular Core:** Scaffold Vite + React repo, implement responsive glassmorphism UI, configure 6-language i18n translation dictionary engine.
- **Hours 4 – 8: Marketplace & Carbon Math:** Build farmer residue listing flow, implement dynamic formula computing ₹ profit + CO₂ tonnes saved, and create buyer quote portal.
- **Hours 9 – 12: Voice AI & AI Plant Doctor:** Integrate Web Speech API for hands-free floating voice nav; implement HTML5 canvas laser scanner for disease diagnosis & remedies.
- **Hours 13 – 15: Mandi Ticker, QA & Live Deploy:** Connect live APMC mandi price feeds, assemble carbon credit leaderboard, run end-to-end testing, and deploy live to Netlify CI/CD.

TECHNOLOGY STACK & TOOLS

- **Frontend UI / UX:** React.js 18, Vite, CSS3, Tailwind, Glassmorphic responsive layouts.
- **Speech & Device APIs:** Web Speech API (SpeechRecognition & Synthesis), MediaDevices API, FileReader API.
- **Computational Logic:** Rule-based carbon offset formulas, GPS proximity filters, APMC mandi datasets.
- **Deployment & DevOps:** Netlify CI/CD, SSL encryption, Progressive Web App (PWA) offline caching.

IMPACT & FUTURE POTENTIAL

Socio-Economic Value Creation, Decarbonization Metrics & Post-Hackathon Vision

1. WHO BENEFITS? [MULTI-STAKEHOLDER VALUE]

- **Smallholder Farmers:** Earn ₹15,000–₹25,000 supplementary income per harvest season from upcycled residue; eliminate middleman commission loss.
- **Industrial Biomass Buyers:** Secure a predictable, traceable supply of agro-waste feedstock for bio-energy, paper pulp, and sustainable packaging at lower haulage costs.
- **Rural Society & Public Health:** Curbs seasonal smog crises and PM2.5 emissions; preserves soil microbial ecology damaged by open field fires.

2. QUANTIFIED IMPACT [MEASURABLE ECOLOGICAL METRICS]

- **10,000+ Tonnes Stubble:** Diverted from open combustion per agricultural cluster annually.
- **15,000+ Tonnes CO₂e:** Greenhouse gas emissions abated, aligning with India's Net-Zero climate mission.
- **+25% to 35% Income Uplift:** Direct economic empowerment injected into rural farming households.
- **100% Linguistic Inclusion:** Zero-literacy voice interface enabling democratic access for all rural demographics.

3. FUTURE VISION & ROADMAP [SCALING BEYOND 15-HOUR MVP]

- **Phase 1 - Field Pilot:** Partner with 5 Farmer Producer Organizations (FPOs) across Karnataka & Punjab to onboard 1,000+ active farmers.
- **Phase 2 - Offline Edge AI:** Deploy quantized lightweight computer vision models (TensorFlow Lite) for 100% offline plant diagnosis in remote fields.
- **Phase 3 - Carbon Finance:** Tokenize verified emissions offsets into certified carbon credits for direct corporate ESG compliance trading.